

ViBridge-VQA: Bridging Global and Local Visual Tokens for Parameter-Efficient Vietnamese Visual Question Answering

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Abstract. Adapting a vision-language model (VLM) to a new Vietnamese Visual Question Answering (VQA) benchmark usually means fine-tuning it on many high-resolution image tiles or training a new architecture. We propose **ViBridge-VQA**, which improves the Vietnamese VLM Vintern-1B while keeping its vision encoder, its projector and its language decoder frozen. ViBridge-VQA feeds the decoder two complementary kinds of visual tokens from a single image view: *global* tokens, produced by a small trainable bridge from the [CLS] vector that Vintern-1B discards, and *local* tokens, produced by Vintern-1B’s own pre-aligned projector and pooled to a chosen budget. Only the global bridge is trained (1.35% of the parameters). We select the token budget as the knee of the accuracy–cost Pareto front, which falls on the same configuration (14 global and 36 local tokens) for all eight metrics on both validation and test, with cost measured in tokens or in inference time. On AutoViVQA, ViBridge-VQA raises the F1 of Vintern-1B from 17.6 to 54.6 and outperforms the officially fine-tuned Vintern-1B on seven of eight metrics while passing 50 instead of up to 1,792 visual tokens; with all 144 local tokens it outperforms it on all eight (F1 +2.3, CIDEr +43.9). Local tokens matter far more than global ones and help most on perception-heavy questions, and they are complementary to decoder adaptation: adding a rank-16 decoder LoRA for one epoch raises F1 to 58.2 and CIDEr to 122.9, the best accuracy and generation scores among all compared systems. Out of distribution, a single view with 144 local tokens exceeds six-tile Vintern-1B on the scene-text ViTextVQA.

Keywords: Vietnamese Visual Question Answering · Vision–Language Models · Parameter-Efficient Fine-Tuning · Low-Rank Adaptation · Generative VQA

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1 Introduction

Visual Question Answering (VQA) is a multimodal task in which a model must interpret an image, understand a natural-language question and produce an answer. Recent progress has been driven by large vision-language models (VLMs) such as BLIP-2 [11], Flamingo [1] and LLaVA [17], which pair a pre-trained vision encoder with a pre-trained language model. For Vietnamese, Vintern-1B [6] follows this design: it connects an InternViT-300M encoder [5] to a Qwen2.5-0.5B decoder [25] through a small projector and is among the strongest open Vietnamese VLMs. Benchmarks such as ViVQA [27], OpenViVQA [20] and, more recently, the reasoning-oriented AutoViVQA [19] allow such models to be evaluated in a low-resource setting.

Obtaining a strong model on a new Vietnamese benchmark, however, remains costly. One option is to fine-tune an existing VLM: the fine-tuned Vintern-1B baseline of AutoViVQA adds LoRA to the language model and encodes every image as up to six high-resolution tiles plus a thumbnail, so that the decoder reads up to 1,792 visual tokens per image. Another option is to design a new architecture, as ViMoE-VQA [18] does with a sparse mixture-of-experts fusion module and a decoder trained on top of frozen encoders. A cheaper alternative is to keep the VLM frozen and train only a small connector, but the choice of connector is delicate. A connector that reads a single global image vector is light but cannot convey small objects, text or positions, whereas connectors that learn to compress the patch tokens, such as Q-Formers or learned pooling, must learn a new alignment to the decoder from limited data. This raises the question: *which visual tokens should a frozen Vietnamese VLM receive, and how many?*

We propose **ViBridge-VQA**, which answers this question with two complementary kinds of tokens. Vintern-1B passes its decoder one token per image region through a projector that was pre-trained to align the two backbones, but it discards the global [CLS] vector of the encoder. ViBridge-VQA reuses that frozen projector to produce *local* tokens, average-pooled to a chosen budget and placed in Vintern-1B’s own image slot, and adds *global* tokens produced from the [CLS] vector by a small trainable bridge. The vision encoder, the projector and the decoder remain frozen; only the global bridge is trained (1.35% of the parameters), and each image is encoded once at 336×336 pixels. Because more tokens cost more computation, we treat the number of global and local tokens as a budget and select it as the knee of the accuracy-cost Pareto front.

Contributions. This work makes the following contributions:

- We propose **ViBridge-VQA**, a global-local bridge that improves Vintern-1B without updating any of its weights. With 14 global and 36 local tokens it raises F1 on AutoViVQA from 17.6 (zero-shot) to 54.6 and outperforms the officially fine-tuned Vintern-1B on seven of eight metrics while passing 50 instead of up to 1,792 visual tokens; with all 144 local tokens it outperforms it on all eight metrics (F1 +2.3, CIDEr +43.9), and with a one-epoch decoder LoRA added it reaches 58.2 F1 and 122.9 CIDEr, the best accuracy and generation scores among all compared systems.

- We study the **visual token budget** over eight configurations and three seeds and select it with an explicit criterion: the knee of the Pareto front falls on the same configuration for all eight metrics, on validation and test, whether cost is counted in tokens or in measured inference time, and a multi-criteria decision analysis (TOPSIS) confirms the choice.
- We provide a **component-wise analysis**: local tokens matter far more than global ones and help most on perception-heavy questions; bridges that learn to compress the patches do not beat a bridge over the global vector; the frozen-decoder model already surpasses every global-only bridge with a decoder LoRA, yet the two are complementary, as local tokens help perception and decoder LoRA reasoning questions. Out of distribution, 144 local tokens let a single view exceed six-tile Vintern-1B on the scene-text ViTextVQA, although Vintern-1B stays better on OpenViVQA.

2 Related Work

Early VQA systems treated the task as classification over a fixed answer set, combining visual features with recurrent or attention-based language models. Transformer-based multimodal models then became the dominant paradigm, and generative VLMs now produce free-form answers with an autoregressive language model [11,1,17].

Vietnamese VQA. Vietnamese VQA resources include ViVQA [27], the open-ended OpenViVQA [20], and specialised datasets for synthetic reasoning (ViCLEVR [29]), book covers (ViOCRQA [24]) and scene text (ViTextVQA [22]). AutoViVQA [19] provides 37,077 LLM-generated and ensemble-filtered questions on 19,411 COCO images [15], each with five short answers and a reasoning label, and emphasises relational, causal and multi-step reasoning. On the modelling side, BARTPhoBEiT [28] and AViVQA-TranConI [21] combine visual and Vietnamese language representations, Vintern-1B [6] brings an instruction-tuned VLM to Vietnamese, and ViMoE-VQA [18] introduces sparse expert routing for generative VQA on AutoViVQA. These approaches either train a new model or fine-tune an existing VLM on multi-tile inputs. We instead keep all pre-trained weights of Vintern-1B frozen and use a single image view.

Frozen backbones and vision-language connectors. Many VLMs keep a pre-trained image encoder and language model frozen and train only a connector between them: Frozen [30] learns a visual prefix, BLIP-2 [11] a Q-Former, and LLaVA [17,16] an MLP projector over all patch tokens. Honeybee [4] argues that connectors should preserve locality, and InternVL [5], on which Vintern-1B is built, merges neighbouring patches before its projector. We compare connectors that read the global vector or learn to compress the patches with one that reuses the pre-trained projector of the VLM itself.

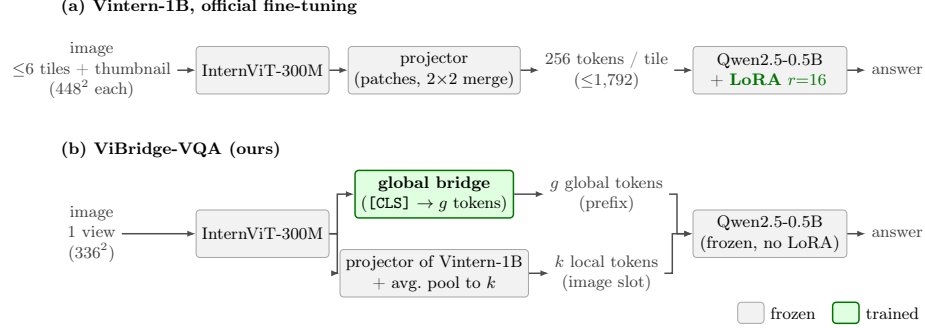


Fig. 1: (a) The official fine-tuning of Vintern-1B encodes up to seven views per image, passes up to 1,792 visual tokens to the decoder and adapts it with LoRA. (b) ViBridge-VQA encodes a single view and passes g global tokens, produced from the [CLS] vector by a trainable bridge, and k local tokens, produced by the frozen projector of Vintern-1B and average-pooled. Only the global bridge is trained (1.35% of the parameters for $g = 14$); we use $g = 14$ and $k = 36$.

Visual token budget. The number of visual tokens dominates the cost of the decoder. Work on inference-optimal VLMs [12] finds that few tokens can suffice for many tasks, and Matryoshka models [3] train one model at several pooled token budgets and show that the required budget depends on the task. We vary the number of global and local tokens separately and select the budget with explicit criteria: the knee of the Pareto front [26] and TOPSIS [10].

Parameter-efficient adaptation. Adapters [8], prefix tuning [13] and LoRA [9] adapt a frozen network by training a small number of additional weights. We compare training a decoder LoRA with giving the frozen decoder richer visual tokens, which shows where a small budget is best spent in a frozen VLM.

3 Proposed Method

We formulate VQA as conditional answer generation: given an image I and a Vietnamese question q , the model generates a short free-form answer $y = (y_1, \dots, y_T)$. ViBridge-VQA starts from Vintern-1B [6], keeps all of its pre-trained weights frozen, and gives its decoder two kinds of visual tokens: global tokens from a small trainable bridge and local tokens from the frozen projector of Vintern-1B. Fig. 1 contrasts this design with the official fine-tuning recipe of Vintern-1B.

3.1 Visual Encoding

The vision encoder is InternViT-300M (24 layers, width 1024) and the language decoder is Qwen2.5-0.5B-Instruct (24 layers, width 896). Each image is resized to a single 336×336 view, which InternViT splits into 24×24 patches of 14×14 pixels. The encoder returns one 1024-dimensional vector per patch and a global [CLS] vector $v \in \mathbb{R}^{1024}$. Vintern-1B discards v ; it merges every 2×2 block of

neighbouring patch vectors into one 4096-dimensional vector (pixel shuffle) and maps each merged vector into the decoder input space with a two-layer MLP projector P that was pre-trained to align the two backbones. At our resolution this yields a 12×12 grid of 144 region tokens.

3.2 Global and Local Tokens

Global tokens. A small trainable bridge g_ϕ expands the [CLS] vector into g tokens with two linear heads:

$$g_\phi(v) = [W_a v + b_a; \text{reshape}_{(g-1) \times 896}(W_e v + b_e)] \in \mathbb{R}^{g \times 896}. \quad (1)$$

The first head produces an *anchor* token and the second $g - 1$ complementary tokens that present the global image content to the frozen decoder from different learned directions. For $g = 14$ the bridge has 12.86M parameters.

Local tokens. The local tokens are the 144 region tokens of the frozen projector P , average-pooled over the 12×12 grid to an $s \times s$ grid, so that $k = s^2$ with $s \in \{1, 3, 6, 12\}$; $k = 144$ keeps every region token and $k = 36$ averages each 2×2 block. Pooling the output of P , rather than learning a new projection of the patches, keeps the local tokens in the representation the decoder was pre-trained to read. The k tokens replace the embeddings of k image-context placeholders in Vintern-1B’s own image slot, $\langle \text{img} \rangle \dots \langle / \text{img} \rangle$, inside the prompt, while the g global tokens are placed before the prompt.

3.3 Training

Only ϕ is trained; the vision encoder, the projector and the decoder remain frozen, and no adapter is added to the decoder. We minimise the answer cross-entropy

$$\mathcal{L} = - \sum_{t=1}^T \log p(y_t \mid y_{<t}, g_\phi(v), \text{pool}_s(P(x)), q), \quad (2)$$

where x are the merged patch vectors. With $g = 14$, 12.86M parameters (1.35%) are trainable. We also train ViBridge-VQA jointly with a rank-16 decoder LoRA [9] on the query, key, value and output projections of every decoder layer (2.16M additional parameters, 15.02M or 1.57% in total), and, for comparison, the *global-only* model ($k = 0$, the Multi-Token bridge with $g = 8$) with and without the same LoRA.

3.4 Choosing the Token Budget

The decoder cost grows with the number of visual tokens $g + k$, so we treat (g, k) as a budget. Accuracy rises with every token step in our experiments, so every configuration lies on the accuracy–cost Pareto front and the front alone does not select one. We therefore pick its *knee* [26]: after normalising cost c and accuracy

a to $[0, 1]$ over the front, the knee maximises $\hat{a} - \hat{c}$, the height above the chord between the cheapest and the most accurate configuration. We compute it on the validation split only and check it with TOPSIS [10] under several weightings of accuracy and cost.

4 Experiments

4.1 Experimental Setup

Dataset. We evaluate on AutoViVQA [19], a Vietnamese VQA benchmark built from MS COCO images [15] with an LLM-driven generation pipeline and ensemble-based quality control. Each question has five short free-form answers (1–10 words) and one of nine question types. The public release is a random 80/20 train/validation split (29,661 / 7,416 questions), in which questions about the same image can fall into different splits. We remove 273 duplicated (image, question) pairs and 97 questions with irregular type labels (77 of them of the rare *comparison* type), keeping eight reasoning categories, and re-partition the remaining 36,707 questions by image into a grouped 70/15/15 split (seed 42), so that no image appears in more than one split: 25,776 / 5,463 / 5,468 questions on 13,579 / 2,909 / 2,910 images for training, validation and test. Compared with a random split, the grouped split lowers F1 by at most 0.5 points; we use it for every number reported here.

Evaluation metrics. Following AutoViVQA and ViMoE-VQA, we report answer-matching metrics (exact-match Accuracy, token Precision, Recall and F1) and generation metrics (BLEU [23], ROUGE-L [14], METEOR [2], CIDEr [31]), all scaled by 100. Token-level scores are computed against each of the five references and the best match is kept. Our F1 averages per-sample F1 scores and is therefore never larger than the harmonic mean $2PR/(P + R)$ of averaged precision and recall, the convention that matches most published baseline values; under that convention ViBridge-VQA would score 55.95 instead of 54.56, so our F1 is conservative.

Implementation details. The vision encoder, the projector and the decoder are initialised from Vintern-1B-v3.5 [6] and kept frozen. Models without a decoder LoRA, including ViBridge-VQA, are trained for 2 epochs, after which validation cross-entropy and CIDEr no longer improve; the decoder-LoRA variants are trained for 1 or 3 epochs. We use AdamW with a learning rate of 2×10^{-4} and an effective batch size of 8, train on the first of the five reference answers, and decode greedily. Every configuration is trained with 3 seeds (42, 123, 3407), and the global-only Multi-Token model additionally with seed 2026. All runs fit on a single 16 GB GPU (Tesla P100 or T4). The token budget is chosen on the validation split; the test split is used only for reporting.

Table 1: Performance on AutoViVQA (test, $\times 100$). Baselines are taken from AutoViVQA [19] and ViMoE-VQA [18]; ViMoE-VQA is the mean over 5 seeds, our models the mean over 3 seeds (global only: 4), with standard deviations in Table 2. g/k : global/local tokens; percentages are trained parameters; ours use 50 ($k=36$) or 158 ($k=144$) visual tokens. Best results per column are in bold. [†]Outside the range of every other system; not used for comparison.

Category	Model	Acc	Prec	Rec	F1	BLEU	ROUGE	METEOR	CIDEr
Multimodal (fine-tuned)	ViT5_ViT	7.97	46.84	50.33	48.52	4.13	46.89	31.02	72.68
	BARTPhoBEiT	8.81	45.30	46.48	45.88	43.29 [†]	44.83	24.57	188.96 [†]
	Vintern-1B	13.01	52.47	55.12	53.76	6.11	51.93	35.25	72.84
	ViMoE-VQA	9.55	62.89	58.60	60.69	12.54	47.12	39.10	88.76
Zero-shot	Vintern-1B	0.12	17.52	19.87	17.55	1.91	25.84	23.93	8.54
	LLaMA 3.2	0.36	23.96	73.71	36.16	3.62	36.11	30.01	62.84
	Gemini 2.0 Flash	0.55	27.20	74.10	39.79	4.41	39.60	31.72	74.42
	Gemini 2.5 Flash	0.22	24.43	76.66	24.75	0.39	37.27	31.22	71.90
	GPT-5	10.84	47.20	55.20	50.89	6.07	47.30	33.34	84.20
Global only	Multi-Token, $g=8$ (0.78%)	8.86	51.19	52.37	50.47	16.24	48.70	41.10	96.19
	+ decoder LoRA, 3 ep. (1.01%)	11.40	55.37	56.16	54.52	20.97	52.68	45.08	108.43
ViBridge-VQA (ours)	$g=14, k=36$ (1.35%)	11.75	55.36	56.56	54.56	18.52	52.64	45.53	110.58
	$g=14, k=144$ (1.35%)	13.19	56.64	58.17	56.04	19.79	54.11	47.38	116.74
	$k=36$ + LoRA, 1 ep. (1.57%)	14.67	59.02	59.96	58.17	22.53	56.25	49.32	122.88

Baselines. We compare against (1) multimodal models fine-tuned on AutoViVQA (ViT5_ViT, BARTPhoBEiT [28] and Vintern-1B [6]), (2) models used zero-shot (Vintern-1B, LLaMA 3.2, Gemini 2.0 and 2.5 Flash, GPT-5), and (3) ViMoE-VQA [18]. Results of groups (1) and (2) are taken from the AutoViVQA benchmark and those of ViMoE-VQA from its paper; both were obtained on an 8:1:1 split of the original release. The fine-tuned Vintern-1B follows the official recipe of Vintern: the vision encoder and the projector are frozen, a rank-16 LoRA is applied to the language model, and each image is encoded as up to six 448×448 tiles plus a thumbnail, for one epoch.

4.2 Main Results

Improvement over Vintern-1B. Table 1 shows that ViBridge-VQA substantially improves the model it is built on without updating any of its weights. With the selected budget of 14 global and 36 local tokens it raises F1 from 17.55 (zero-shot) to 54.56 and CIDEr from 8.54 to 110.58. It outperforms the officially fine-tuned Vintern-1B on seven of the eight metrics, precision (+2.9), recall (+1.4), F1 (+0.8), BLEU (+12.4), ROUGE-L (+0.7), METEOR (+10.3) and CIDEr (+37.7), and is lower only in exact-match accuracy (−1.3), while passing 50 visual tokens to the decoder instead of up to 1,792 and encoding one view instead of up to seven. With all 144 local tokens it outperforms the fine-tuned Vintern-1B on all eight metrics, including accuracy (+0.2), F1 (+2.3), METEOR (+12.1) and CIDEr (+43.9), still with 11 times fewer visual tokens.

Adding decoder adaptation. Training the selected budget jointly with a rank-16 decoder LoRA for one epoch (15.02M trained parameters, 1.57%) adds 3.6 F1 on test and gives the best model: F1 58.17, CIDEr 122.88 and accuracy 14.67, above

Table 2: Our models on the validation and test splits (mean $\pm$ std over seeds: global only 4 seeds, the others 3; $\times 100$). CE is the answer cross-entropy on the split. Best value per split in bold.

Split	Config.	Acc	Prec	Rec	F1	BLEU	ROUGE-L	METEOR	CIDEr	CE
val	global only	9.21 $\pm$.05	51.54 $\pm$.15	52.48 $\pm$.21	50.74 $\pm$.17	16.88 $\pm$.23	49.03 $\pm$.12	41.38 $\pm$.21	99.07 $\pm$.47	1.491
	+ LoRA, 3 ep.	12.04 $\pm$.15	55.58 $\pm$.25	56.38 $\pm$.23	54.78 $\pm$.22	21.25 $\pm$.03	53.05 $\pm$.15	45.46 $\pm$.22	110.67 $\pm$.48	1.327
	ours, $k=36$	12.38 $\pm$.22	55.86 $\pm$.21	56.72 $\pm$.33	54.96 $\pm$.28	18.73 $\pm$.23	53.10 $\pm$.27	46.01 $\pm$.42	113.29 $\pm$ 1.01	1.396
	ours, $k=144$	14.19 $\pm$.36	57.36 $\pm$.07	58.58 $\pm$.03	56.61 $\pm$.04	20.18 $\pm$.13	54.70 $\pm$.05	48.00 $\pm$.05	120.71 $\pm$.12	1.365
	ours, $k=36$ + LoRA	14.82 $\pm$.26	59.08 $\pm$.31	59.93 $\pm$.07	58.21 $\pm$.16	22.27 $\pm$.17	56.42 $\pm$.14	49.55 $\pm$.06	124.88 $\pm$.75	1.267
test	global only	8.86 $\pm$.37	51.19 $\pm$.19	52.37 $\pm$.21	50.47 $\pm$.22	16.24 $\pm$.43	48.70 $\pm$.19	41.10 $\pm$.29	96.19 $\pm$ 1.10	1.500
	+ LoRA, 3 ep.	11.40 $\pm$.24	55.37 $\pm$.18	56.16 $\pm$.09	54.52 $\pm$.14	20.97 $\pm$.29	52.68 $\pm$.21	45.08 $\pm$.14	108.43 $\pm$.67	1.329
	ours, $k=36$	11.75 $\pm$.21	55.36 $\pm$.21	56.56 $\pm$.21	54.56 $\pm$.19	18.52 $\pm$.18	52.64 $\pm$.19	45.53 $\pm$.33	110.58 $\pm$.95	1.412
	ours, $k=144$	13.19 $\pm$.23	56.64 $\pm$.13	58.17 $\pm$.26	56.04 $\pm$.19	19.79 $\pm$.15	54.11 $\pm$.16	47.38 $\pm$.26	116.74 $\pm$.76	1.377
	ours, $k=36$ + LoRA	14.67 $\pm$.30	59.02 $\pm$.12	59.96 $\pm$.33	58.17 $\pm$.19	22.53 $\pm$.11	56.25 $\pm$.22	49.32 $\pm$.37	122.88 $\pm$.93	1.276

the fine-tuned Vintern-1B on all eight metrics (F1 +4.4, BLEU +16.4, METEOR +14.1, CIDEr +50.0) and above the global-only model with the same one-epoch LoRA by 5.1 F1. Local tokens and decoder adaptation are thus complementary rather than interchangeable (Section 4.5).

Comparison with other systems. ViBridge-VQA with a decoder LoRA achieves the best accuracy, BLEU, ROUGE-L, METEOR and CIDEr among all compared systems. ViMoE-VQA, a task-specific model whose fusion module and decoder are trained on AutoViVQA, keeps the highest precision and F1 (2.5 F1 above ours). Compared with zero-shot large language models, ViBridge-VQA is better on every metric except recall: zero-shot models produce long answers that cover many reference words at the cost of precision, whereas ViBridge-VQA produces concise answers with a balanced precision–recall trade-off. Even with a frozen decoder, ViBridge-VQA matches the global-only model trained with a 3-epoch decoder LoRA (54.56 versus 54.52 F1) and surpasses it with $k = 144$.

Stability. All configurations are stable across random seeds (Table 2): the standard deviation of ViBridge-VQA is at most 0.28 for F1 and 1.01 for CIDEr, and test F1 lies at most 0.6 points below validation F1, so the budget chosen on the validation split does not over-fit it.

4.3 Choosing the Token Budget

Table 3 and Fig. 2a vary the number of global tokens $g \in \{8, 14\}$ and local tokens $k \in \{1, 9, 36, 144\}$. *Local tokens matter far more than global ones:* at $g = 14$, going from 1 to 144 local tokens adds 4.9 F1 and 19.1 CIDEr points on validation, whereas going from 8 to 14 global tokens adds 0.2–0.6 F1 at a fixed k . Accuracy does not saturate within the 144 region tokens available at our resolution: each step of k in Table 3 adds 1.3–2.0 F1. The gain per added token, however, falls quickly, from 0.07 F1 per token between 23 and 50 tokens to 0.015 between 50 and 158, so the budget is a genuine trade-off.

Since every configuration lies on the Pareto front, we select the knee (Section 3.4). On the validation split it is $g = 14$, $k = 36$, and the same configuration is the knee for each of the eight metrics on both validation and test, both when cost is counted in visual tokens and when it is measured as inference time; with

Table 3: Token budget of ViBridge-VQA ($\times 100$, mean $\pm$ std over 3 seeds; global only: 4 seeds). Tokens: visual tokens passed to the decoder ($g + k$). Time: wall-clock minutes of one evaluation session on the 10,931 validation and test questions, including model loading (mean over the 3 seeds; Kaggle P100/T4). The selected budget is in bold.

g	k	Tokens	Time	F1 (val)	CIDEr (val)	F1 (test)	CIDEr (test)
8	0	8	—	50.74 $\pm$.17	99.07 $\pm$.47	50.47 $\pm$.22	96.19 $\pm$ 1.10
8	1	9	57.0	51.03 $\pm$.12	99.27 $\pm$.58	50.71 $\pm$.16	96.90 $\pm$.87
8	9	17	59.2	52.82 $\pm$.11	106.11 $\pm$.55	52.11 $\pm$.25	101.30 $\pm$.87
8	36	44	60.6	54.32 $\pm$.41	111.88 $\pm$ 1.43	53.82 $\pm$.22	107.67 $\pm$.53
8	144	152	71.2	56.15 $\pm$.08	119.05 $\pm$.25	55.20 $\pm$.07	114.13 $\pm$.46
14	1	15	58.8	51.67 $\pm$.25	101.66 $\pm$ 1.11	51.67 $\pm$.12	99.53 $\pm$.93
14	9	23	57.7	52.98 $\pm$.08	106.02 $\pm$.80	52.76 $\pm$.28	103.63 $\pm$ 1.24
14	36	50	60.0	54.96 $\pm$.28	113.29 $\pm$ 1.01	54.56 $\pm$.19	110.58 $\pm$.95
14	144	158	70.2	56.61 $\pm$.04	120.71 $\pm$.12	56.04 $\pm$.19	116.74 $\pm$.76

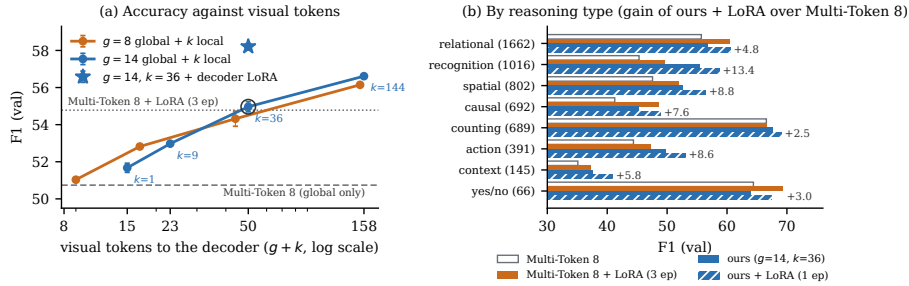


Fig. 2: (a) Validation F1 against the number of visual tokens (mean $\pm$ std over 3 seeds); the circled point is the knee, $g = 14$, $k = 36$. Horizontal lines: the global-only Multi-Token model (4 seeds) and the same model with a 3-epoch decoder LoRA (3 seeds). (b) Validation F1 by reasoning type (number of questions in parentheses): global only and + LoRA for seed 42, ViBridge-VQA ($g = 14$, $k = 36$) without and with a 1-epoch decoder LoRA averaged over 3 seeds; numbers are the gain of ViBridge-VQA + LoRA over the global-only model.

a logarithmic token axis it is the knee in 11 of 16 cases. As a multi-criteria check, TOPSIS with the gains over the global-only model in F1, CIDEr and METEOR as benefits and the added tokens, inference time, trained parameters and seed standard deviation as costs ranks this configuration first when accuracy and cost are weighted equally; weighting cost more favours $k = 9$ and weighting accuracy more favours $k = 144$, but $g = 14$, $k = 36$ is the only configuration among the top four for every accuracy weight from 0.3 to 0.7. It is also the cheapest configuration that reaches the global-only model with a 3-epoch decoder LoRA (54.96 versus 54.78 validation F1), and it obtains 72% of the largest F1 gain (4.2 of 5.9 points) with 32% of the tokens and 15% less inference time than $k = 144$. We use 14 global tokens because they are better than 8 at every k (by 0.5–0.6 F1 at $k = 1$, 36 and 144, and by 0.2 at $k = 9$), in line with the earlier observation that the global-only bridge plateaus at 14 tokens (Section 4.5).

4.4 Efficiency

ViBridge-VQA trains 12.86M parameters, encodes one view per image and passes 50 visual tokens to the decoder. On a Tesla P100, encoding one 336×336 view with InternViT-300M costs 362 GFLOPs and 229 ms, whereas the six 448×448 tiles used by the fine-tuned Vintern-1B (plus a thumbnail) cost 2,172 GFLOPs and 922 ms, and give the decoder up to 1,792 visual tokens, 36 times more than ViBridge-VQA. Within our own budgets, keeping all 144 local tokens instead of 36 makes an evaluation session about 17% slower (Table 3).

4.5 Ablation Study

All ablations are reported on the validation split.

Global-only bridges and decoder adaptation. Table 4 compares five bridges that do not use the projector of Vintern-1B: a *Residual* bridge producing one token, the *Multi-Token* bridge over the [CLS] vector, a *Tile-Attention* bridge in which 8 learned queries attend over the patch tokens, and a *Light* and a *Full* Q-Former [11] with 8 and 16 queries. Without decoder adaptation they span 45.14–50.74 F1, and the Multi-Token bridge is the best: the bridges that learn to compress the patches, with up to $9.4\times$ more parameters, are 3.7–5.6 F1 lower, so learning a new alignment of the patch tokens from 25k questions does not pay off. A decoder LoRA improves every bridge and brings them within 0.8 F1 (52.69–53.47). ViBridge-VQA takes a different route: it keeps the decoder frozen but gives it local tokens that are already aligned with it, and reaches 54.96 F1 with $k = 36$ and 56.61 with $k = 144$, above every bridge with a decoder LoRA. Decoder adaptation is therefore not needed to surpass them, but it remains complementary: in the same one-epoch joint run it adds 3.3 F1 to the global-local bridge (58.21), more than the 2.7 it adds to the global-only one. Two further interventions on the global-only bridge confirm that the projector’s information must be passed as tokens rather than distilled: training the bridge towards the projector of Vintern-1B, at the feature level or on the decoder’s output distribution, lowers F1 by 0.39 and 0.28 (and by 9.05 when the distillation term dominates), and sampling a random reference answer per epoch lowers it by 2.07.

Bridge design. In an earlier evaluation setting (generation read a 448×448 crop), increasing the global-only bridge from 4 to 20 output tokens raised F1 up to 14 tokens (50.67) and then plateaued (50.44–50.62), which motivates $g = 14$. Replacing the global vector with bridges that pool the raw patch tokens through a newly learned projection loses accuracy: fixed mean and max pooling lose 10.8 and 13.9 F1, learned attention loses 5.1 F1, and a Honeybee-style convolutional abstractor [4] comes within 0.7 F1 with $2.7\times$ the parameters. Pooling the output of the pre-trained projector instead, as ViBridge-VQA does, turns the same kind of operator into a gain: with the same 8 global tokens, adding the region tokens averaged over 2×2 blocks ($k = 36$) raises F1 by 3.6 points (Table 3).

Table 4: Global-only bridges before and after decoder LoRA (validation, $\times 100$; mean $\pm$ std over 3 seeds, Multi-Token bridge only 4 seeds; *seed 42 only). Params: bridge parameters (LoRA adds 2.16M). Δ F1: gain of + LoRA over bridge only. Best value per group in bold.

Bridge	k	Params	Acc	Prec	Rec	F1	BLEU	ROUGE-L	METEOR	CIDEr	CE	Δ F1
<i>Bridge only (2 epochs)</i>												
Residual	1	4.86M	5.63 $\pm$.59	46.31 $\pm$.34	48.79 $\pm$.33	46.02 $\pm$.36	12.98 $\pm$.15	44.23 $\pm$.38	36.86 $\pm$.31	86.48 $\pm$.77	1.665	–
Tile-Attn.	8	4.14M	4.99 $\pm$ 1.10	45.49 $\pm$ 1.10	48.05 $\pm$.60	45.14 $\pm$.92	12.34 $\pm$.32	43.28 $\pm$.95	36.11 $\pm$.73	84.14 $\pm$ 1.40	1.683	–
Multi-Token	8	7.35M	9.21 $\pm$.05	51.54 $\pm$.15	52.48 $\pm$.21	50.74 $\pm$.17	16.88 $\pm$.23	49.03 $\pm$.12	41.38 $\pm$.21	99.07 $\pm$.47	1.491	–
Light Q-F.	8	27.6M	6.94 $\pm$.32	47.74 $\pm$.80	49.18 $\pm$.51	47.06 $\pm$.66	13.99 $\pm$.33	45.37 $\pm$.63	37.57 $\pm$.60	88.07 $\pm$ 1.89	1.599	–
Full Q-F.	16	69.4M	6.97 $\pm$.33	47.42 $\pm$.70	49.02 $\pm$.35	46.84 $\pm$.47	13.76 $\pm$.36	45.10 $\pm$.41	37.43 $\pm$.34	87.74 $\pm$ 1.24	1.605	–
<i>+ LoRA (joint, 1 epoch)</i>												
Residual	1	4.86M	10.36 $\pm$.10	53.51 $\pm$.06	54.36 $\pm$.34	52.69 $\pm$.14	18.94 $\pm$.36	51.04 $\pm$.13	43.33 $\pm$.33	104.25 $\pm$.93	1.396	+6.67
Tile-Attn.*	8	4.14M	10.34	53.72	54.43	52.82	19.13	51.25	43.44	104.72	1.400	+7.68
Multi-Token	8	7.35M	10.94 $\pm$.12	54.37 $\pm$.11	55.05 $\pm$.22	53.47 $\pm$.17	19.72 $\pm$.09	51.79 $\pm$.15	44.11 $\pm$.23	106.68 $\pm$.69	1.374	+2.73
Light Q-F.	8	27.6M	10.76 $\pm$.12	54.06 $\pm$.16	54.78 $\pm$.17	53.21 $\pm$.14	19.92 $\pm$.31	51.54 $\pm$.13	43.77 $\pm$.19	106.26 $\pm$.73	1.378	+6.14
Full Q-F.	16	69.4M	10.81 $\pm$.15	54.10 $\pm$.41	54.91 $\pm$.23	53.29 $\pm$.29	19.57 $\pm$.25	51.62 $\pm$.30	43.96 $\pm$.21	106.50 $\pm$ 1.04	1.372	+6.45

4.6 Gains by Reasoning Type

Fig. 2b breaks the results down by the reasoning categories of AutoViVQA. The local tokens help most where the answer depends on what is visible in the image: recognition (+10.2 F1 over the global-only model), action (+5.4) and spatial (+5.1) questions, on which the frozen-decoder ViBridge-VQA is also better than the global-only model with a decoder LoRA. That model is better on causal and relational questions, whose reference answers are the longest (5.0 and 5.6 words on average), and on yes/no questions. Adding the LoRA to ViBridge-VQA combines both strengths: it is the best model in seven of the eight categories, reaching the decoder-LoRA model on causal (48.9 versus 48.6) and relational (60.5 versus 60.5) questions while keeping the gains of the local tokens on recognition (58.7 versus 49.7). Local tokens improve what the model perceives and decoder adaptation how it phrases reasoning answers.

4.7 Generalisation to Other Datasets

Table 5 evaluates the models zero-shot on four Vietnamese VQA datasets not used in training: two scene-text datasets, ViTextVQA [22] and OpenViVQA [20], and two general-domain ones, ViVQA-X and ViVQA. Local tokens change the picture most on scene text, where the global-only model collapses: on ViTextVQA, F1 rises from 3.96 to 15.14 with 36 local tokens and to 33.26 with 144, which exceeds the zero-shot Vintern-1B reading six 448×448 tiles (32.01) while ViBridge-VQA reads one 336×336 view. On OpenViVQA, whose references are long descriptive sentences, local tokens narrow the gap to Vintern-1B from 36.8 to 22.5 F1, but Vintern-1B remains clearly better. On the general-domain datasets ViBridge-VQA is better than both other models on every metric except recall, by 10.4 (ViVQA-X) and 18.8 (ViVQA) F1 over Vintern-1B with $k = 36$. As on AutoViVQA, the zero-shot model reaches a high recall with long answers but a low precision. Adding the decoder LoRA to $k = 36$ helps on the general-domain datasets (F1 +2.7 on ViVQA-X and +2.1 on ViVQA, close to $k = 144$) but hardly on scene text (+0.3 and +0.8). Reading text thus needs more local tokens, which decoder adaptation cannot replace, consistent with the token-budget curve.

Table 5: Generalisation to Vietnamese VQA datasets not seen in training ($\times 100$; about 1,000 sampled test questions per seed, 505–519 for ViVQA; mean $\pm$ std over 3 sampling seeds). Vintern-1B is used zero-shot with 6 tiles; the other models are trained on AutoViVQA only (seed 42); + LoRA: one-epoch decoder LoRA. Best value per dataset in bold.

Dataset	Model	Acc	Prec	Rec	F1	BLEU	ROUGE-L	METEOR	CIDEr
ViTextVQA (scene text)	Vintern-1B, zero-shot	0.40 $\pm$.08	23.05 $\pm$.17	64.56 $\pm$ 1.26	32.01 $\pm$.31	4.94 $\pm$.18	20.89 $\pm$.25	26.50 $\pm$.54	195.62 $\pm$ 2.35
	global only + LoRA	0.07 $\pm$.05	4.62 $\pm$.16	4.07 $\pm$.12	3.96 $\pm$.09	0.15 $\pm$.01	3.74 $\pm$.12	2.61 $\pm$.16	15.64 $\pm$.92
	ours, $k=36$	3.40 $\pm$.43	17.85 $\pm$.44	14.82 $\pm$.52	15.14 $\pm$.47	1.01 $\pm$.10	14.27 $\pm$.43	10.49 $\pm$.36	66.39 $\pm$ 2.51
	ours, $k=144$	13.90 $\pm$ 1.24	34.26 $\pm$.61	35.52 $\pm$.96	33.26 $\pm$.67	6.24 $\pm$.48	31.65 $\pm$.73	27.33 $\pm$.93	190.30 $\pm$ 6.71
	ours, $k=36$ + LoRA	3.20 $\pm$.24	17.22 $\pm$.65	15.82 $\pm$.07	15.39 $\pm$.39	1.03 $\pm$.15	14.41 $\pm$.33	11.17 $\pm$.26	70.07 $\pm$ 1.33
OpenViVQA (scene text)	Vintern-1B, zero-shot	0.07 $\pm$.05	53.90 $\pm$.84	72.26 $\pm$ 1.01	58.40 $\pm$.80	22.36 $\pm$.82	48.83 $\pm$.90	51.24 $\pm$ 1.06	411.63 $\pm$ 11.11
	global only + LoRA	0.73 $\pm$.21	33.32 $\pm$.56	17.73 $\pm$.49	21.56 $\pm$.48	1.67 $\pm$.15	20.98 $\pm$.52	13.08 $\pm$.37	83.21 $\pm$ 2.37
	ours, $k=36$	2.50 $\pm$.30	44.41 $\pm$ 1.02	23.07 $\pm$.74	28.24 $\pm$.79	1.34 $\pm$.28	27.54 $\pm$.76	18.51 $\pm$.67	128.36 $\pm$ 4.40
	ours, $k=144$	4.30 $\pm$.08	54.06 $\pm$ 1.06	30.17 $\pm$.74	35.89 $\pm$.80	2.77 $\pm$.30	34.64 $\pm$.78	25.06 $\pm$.68	187.00 $\pm$ 5.08
	ours, $k=36$ + LoRA	2.00 $\pm$.30	44.11 $\pm$.73	23.99 $\pm$.38	28.99 $\pm$.51	1.41 $\pm$.12	28.19 $\pm$.50	18.99 $\pm$.33	130.46 $\pm$ 2.43
ViVQA-X [7]	Vintern-1B, zero-shot	0.20 $\pm$.08	9.13 $\pm$.35	61.15 $\pm$.80	14.52 $\pm$.47	0.06 $\pm$.01	8.98 $\pm$.32	11.96 $\pm$.41	50.13 $\pm$ 2.83
	global only + LoRA	2.57 $\pm$.09	13.19 $\pm$.49	26.80 $\pm$ 1.24	15.89 $\pm$.65	0.18 $\pm$.02	11.94 $\pm$.49	9.08 $\pm$.57	42.12 $\pm$ 2.92
	ours, $k=36$	7.40 $\pm$.28	20.68 $\pm$.16	41.32 $\pm$.26	24.91 $\pm$.14	0.34 $\pm$.12	19.38 $\pm$.25	16.00 $\pm$.21	79.51 $\pm$ 1.15
	ours, $k=144$	8.41 $\pm$.37	23.09 $\pm$.12	46.68 $\pm$.18	27.89 $\pm$.13	0.39 $\pm$.13	22.02 $\pm$.26	18.55 $\pm$.11	92.22 $\pm$ 2.31
	ours, $k=36$ + LoRA	8.11 $\pm$.33	23.07 $\pm$.18	44.57 $\pm$.06	27.64 $\pm$.11	0.33 $\pm$.05	21.98 $\pm$.13	19.24 $\pm$.14	95.29 $\pm$ 1.51
ViVQA [27]	Vintern-1B, zero-shot	0.00 $\pm$.00	16.01 $\pm$.18	60.80 $\pm$ 1.17	23.90 $\pm$.24	0.27 $\pm$.02	15.78 $\pm$.25	17.84 $\pm$.19	78.63 $\pm$ 2.07
	global only + LoRA	7.38 $\pm$.53	31.67 $\pm$.97	34.01 $\pm$ 1.58	31.48 $\pm$ 1.15	0.25 $\pm$.09	31.48 $\pm$ 1.16	21.92 $\pm$.82	84.94 $\pm$ 2.87
	ours, $k=36$	16.59 $\pm$ 1.48	40.59 $\pm$ 1.37	48.86 $\pm$ 1.89	42.68 $\pm$ 1.52	0.37 $\pm$.00	42.65 $\pm$ 1.55	34.62 $\pm$ 1.37	171.21 $\pm$ 6.06
	ours, $k=144$	20.17 $\pm$.91	43.12 $\pm$.66	51.09 $\pm$ 1.67	44.95 $\pm$.90	0.59 $\pm$.17	44.95 $\pm$.90	37.28 $\pm$.61	191.28 $\pm$ 2.17
	ours, $k=36$ + LoRA	18.93 $\pm$ 1.37	43.08 $\pm$.93	50.88 $\pm$ 1.71	44.76 $\pm$ 1.21	0.63 $\pm$.02	44.74 $\pm$ 1.22	36.98 $\pm$ 1.28	183.60 $\pm$ 7.17



Fig. 3: Qualitative examples from the AutoViVQA validation set (seed 42). Q: question, GT: one of the five reference answers, Global: Multi-Token model without local tokens, +LoRA: the same model with a 3-epoch decoder LoRA, Ours: ViBridge-VQA ($g = 14$, $k = 36$). Plausible answers are green, wrong ones red.

4.8 Qualitative Analysis

Fig. 3 shows representative predictions. In Examples 1–3 only ViBridge-VQA is right, and each answer depends on a local detail (the brand printed on the laptop, the canal that identifies Venice, the water sprayed by the elephant), where

the global-only models answer “Apple”, “Hanoi” or “swimming”. Example 4 is a failure: ViBridge-VQA describes the scene instead of the bus’s purpose, which both global-only models infer, in line with the advantage of decoder adaptation on causal questions.

5 Conclusion

This work presents **ViBridge-VQA**, which improves the Vietnamese VLM Vintern-1B for generative visual question answering without updating any of its weights. ViBridge-VQA passes the frozen decoder two kinds of visual tokens from a single image view: global tokens, produced from the [CLS] vector by a small trainable bridge, and local tokens, produced by the frozen pre-trained projector of Vintern-1B and pooled to a chosen budget. Selecting that budget as the knee of the accuracy–cost Pareto front gives 14 global and 36 local tokens for every metric and both evaluation splits. With 1.35% of the parameters trained and 50 visual tokens, it raises the F1 of Vintern-1B on AutoViVQA from 17.6 to 54.6 and outperforms the officially fine-tuned Vintern-1B on seven of eight metrics; with all 144 local tokens it outperforms it on all eight. Our analysis shows that local tokens matter far more than global ones, that the pre-trained projector should provide them rather than a newly learned compression of the patches, and that a frozen decoder with such tokens surpasses a decoder adapted with LoRA over global tokens only. The two are nevertheless complementary: adding a one-epoch decoder LoRA to the selected budget gives 58.2 F1 and 122.9 CIDEr, the best accuracy and generation scores among the compared systems.

ViMoE-VQA, a model trained specifically for AutoViVQA, still achieves a higher F1, and the baselines were reported on a different split of AutoViVQA. Out of distribution, the local tokens let a single 336×336 view match the six-tile Vintern-1B on ViTextVQA, but Vintern-1B remains clearly better on OpenViVQA, decoder adaptation does not help on scene text, and accuracy had not saturated at the 144 region tokens available at our resolution. Future work will therefore raise the image resolution. The code and the grouped split are available at <https://github.com/nguyennn263/modular-vlm-finetune>.

Overall, ViBridge-VQA shows that an existing Vietnamese VLM can be improved substantially by choosing which visual tokens its frozen decoder receives, and that the size of this budget can be chosen with an explicit accuracy–cost criterion. We hope this work will encourage efficient and well-evaluated multimodal models for low-resource languages.

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